

Shao Wang

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Research Interests

My research interests lie at the intersection of machine learning and computer systems. I'm currently focusing on building efficient and large-scale systems for emerging ML workloads like LLMs and AI agents.

Education

Shanghai Jiao Tong University | *Computer Science M.S.* 2024.09 - 2027.03 (exp.)

- Cumulative GPA: 3.81/4.00
- Award: Optiver Outstanding Scholarship, First-Class Academic Scholarship

Shanghai Jiao Tong University | *Software Engineering B.Eng.* 2020.09 - 2024.06

- Cumulative GPA: 3.76/4.30
- Award: Huawei Scholarship, Wenyuan Pan Scholarship

Research Experiences

Catalyst Lab, Carnegie Mellon University | *Research Assistant* 2026.04 - Now

- Advisor: Prof. Zhihao Jia.
- Maintaining **Mirage Persistent Kernel (MPK)** (accepted to OSDI'26), a megakernel-based LLM serving system that fuses an entire inference graph into a single persistent GPU kernel for low-latency LLM inference.
- Led the bring-up of MPK's online serving stack, contributed bug fixes, and built the CI/CD pipeline.
- Co-leading **Tangram**, an ongoing megakernel compiler that generates provably optimal megakernels for LLM inference.

Data Service Lab, Shanghai Jiao Tong University 2025.07 - 2026.04

- Advisors: Prof. Lin Gui and Rui Ren.
- Initiated **ForkKV** project, which alleviates the memory footprint bottleneck in multi-LoRA agent serving by enabling efficient KV cache sharing across agents. Achieves up to 3× the throughput of SOTA baselines.
- **Independently led the project** from conception to implementation and manuscript composition.

Publications & Talks

ForkKV: Scaling Multi-LoRA Agent Serving with Copy-on-Write Disaggregated KV Cache 2026.04

- *Shao Wang, Rui Ren, Lin Gui*
- arxiv.org/abs/2604.06370

RAG and Fine Tuning with Kubeflow 2025.11

- *Francisco Javier Arceo, Shao Wang*
- *Talk at KubeCon + CloudNativeCon NA 2025*

Streamline LLM Fine-tuning on Kubernetes with Kubeflow LLM Trainer 2025.11

- *Shao Wang, Andrey Velichkevich*
- *Talk at Kubeflow Summit NA 2025*

Open Source Projects

Mirage Persistent Kernel (MPK) | *Maintainer* 2026.04 - Now

- **MPK** is a megakernel-based LLM serving system that fuses an entire inference graph into a single persistent GPU kernel for low-latency LLM inference. (MPK has earned **2.4k+** stars on GitHub, and is accepted to OSDI'26)
- **Online Serving:** Implemented online serving on MPK's persistent kernel. Since offline mode runs a fixed batch decided at launch, serving live traffic required updating the always-running kernel's batch on the fly; I added a CPU-GPU ring buffer that feeds in new requests and retires finished ones, enabling dynamic batch filling and continuous batching.
- **CI/CD:** Designed and built MPK's CI pipeline on self-hosted B200 (SM100) GPU runners, covering offline token-level correctness/performance, per-batch-size perf tracking, and end-to-end online serving.

Kubeflow | WG AutoML/Training Maintainer

2024.10 - Now

- (Kubeflow has earned **33.2k+** stars on Github and **268M+** downloads on PyPI over the years)
- **LLMOps:** Led the design and implementation of Kubeflow LLM Trainer V2, which streamlines LLM fine-tuning on Kubernetes by: 1) Hiding complex Kubernetes configurations from data scientists via designing CRDs for different personas 2) Unleash data scientists from the burden of manually setting up environment by providing editable workflow templates and a simple yet efficient Python API
- **Mentorship:** Acted as a Mentor for several Kubeflow GSoC'25 projects, leading a three-member team and coordinating cross-time-zone collaboration to support students in accomplishing their project goals.
- **Maintenance:** Participate in the daily maintenance routine for *kubeflow/trainer*, *kubeflow/katib*, and *kubeflow/sdk*

Industry Experiences

Databend Labs | Cloud Platform Intern

2024.10 - 2025.04

- **Operator:** Developed an open-source operator for databend on Kubernetes from scratch, providing open-source users with a flexible yet efficient approach to orchestrate databend clusters on Kubernetes.
- **Observability:** Integrated Prometheus for enhanced system visibility on the on-premise cloud platform.

Goolge Summer of Code | Contributor, CNCF - Kubeflow

2024.05 - 2024.09

- **Push-based Metrics Collection:** Built a push-based training metrics system to replace Katib's sidecar approach, which reduced performance/resource overhead and improved metrics collection latency.
- **SDK:** Developed a simple Python API for metric reporting, minimizing format errors and improving robustness.
- **E2E Testing:** Added e2e tests for hyperparameter tuning in GitHub Actions to enhance CI coverage.

Bondi Tech Ltd. | Infrastructure Intern

2024.01 - 2024.04

- **Investigation:** Evaluated time series databases for OLAP analysis of market data and delivered benchmark reports.
- **Quote Storage:** Build a new storage system from scratch on Kubernetes in replace of the current CSV-based storage system, which enhances the performance of data analysis business by 10x times.